Third Normal Form (3NF) Justification Report

Dataset Overview

The Olist e-commerce dataset comprises comprehensive data on online transactions, including customer details, seller information, product dimensions, order status, and customer reviews. Initially structured as flat, denormalized CSV files, the dataset contained significant redundancy, such as repetitive customer locations and product categories. The objective of this design is to translate these disjointed files into a robust relational database schema that enforces data integrity, eliminates redundancy, and complies with Third Normal Form (3NF).

Core Entities

To organize the raw data into a relational structure, the dataset was decomposed into primary entities representing distinct business concepts. The core entities identified are:

- **`customers` and `sellers`**: Represent the users participating in transactions. Both require location tracking.
- **`locations`**: A centralized repository defining geographical areas (city and state) based on ZIP codes.
- **`products` and `product_categories`**: Isolate item attributes and catalog groupings.
- **`orders`**: The central transactional entity recording the lifecycle timestamps and status of a purchase.
- **`order_payments` and `order_reviews`**: Detail entities capturing financial transactions and customer feedback associated with an order.

Relationships and Bridge Tables

Orders act as the focal point connecting customers to products. Since an order can contain multiple products, and a given product can appear across numerous orders, this inherently creates a many-to-many relationship.

To resolve this, the associative bridge table `order_items` was introduced. This table effectively decomposes the many-to-many relationship into two one-to-many relationships (`orders` to `order_items`, and `products` to `order_items`). Each row in `order_items` represents a specific line item in a purchase, uniquely identified by a composite key (`order_id`, `order_item_id`). It also accommodates transaction-specific facts, such as the `price` and `freight_value`, which depend precisely on the item within the context of that specific order rather than the product universally.

Normalization to 1NF, 2NF, and 3NF

First Normal Form (1NF)

The schema satisfies 1NF by ensuring all attributes are atomic and indivisible. Denormalized, compound strings (such as full addresses) were split into atomic components like

`zip_code_prefix`, `city`, and `state`. Furthermore, there are no repeating groups; for example, multiple items in an order are not stored as arrays but as distinct rows in `order_items`.

Second Normal Form (2NF)

To achieve 2NF, the schema must be in 1NF and contain no partial dependencies. This means all non-key attributes must depend on the *entire* primary key. In a composite key table like `order_items(order_id, order_item_id)`, attributes such as `price` and `seller_id` depend on the exact line item (the combination of both keys), not just on the `order_id` or the `product_id`. Single-key tables implicitly satisfy 2NF since they possess no composite keys.

Third Normal Form (3NF)

A schema reaches 3NF when it is in 2NF and exhibits no transitive dependencies (i.e., non-key attributes must not depend on other non-key attributes). A crucial example of transitive dependency removal in this design is the handling of geographical data. Initially, a customer's `city` and `state` depended transitively on the customer's `zip_code_prefix`. Instead of storing `city` and `state` within `customers` or `sellers`, they were relocated to the independent `locations` table. Consequently, `customers` and `sellers` reference the `zip_code_prefix` via a foreign key. Thus, every non-key attribute in the schema depends exclusively on its respective primary key.

Avoidance of Data Anomalies

The 3NF design directly prevents common data anomalies:

- **Insert Anomalies:** A new geographical location or product category can be inserted independently into the `locations` or `product_categories` master tables without requiring an associated customer or product.
- **Update Anomalies:** Adjusting an English translation for a category name only requires updating a single record in `product_categories`. The update instantly cascades logically to all related products via foreign key relations, rather than risking inconsistent spelling by updating thousands of denormalized product records.
- **Delete Anomalies:** Deleting a customer record does not result in the unintentional loss of the `city` and `state` information, as those facts are preserved safely in the `locations` table.

Design Assumptions or Notes

- **Pricing Dynamics**: It is assumed that the `price` and `freight_value` fluctuate over time or vary by seller. Therefore, these attributes are stored in the `order_items` bridge table at the time of purchase rather than as static values in the `products` table.
- **Location Normalization**: The model assumes that `zip_code_prefix` is a sufficient primary key uniquely identifying a `city` and `state` combination within this Brazilian dataset.
- **Orphan Prevention**: Constraints prevent orphan records. For instance, `NOT NULL` mandates ensure every order belongs to a valid customer, preventing incomplete transaction records.